

Institute of Space Technology	Form No	IST-Dean-F-18/00
Project Proposal Form	Issue Date	
Grading Policy on Final Year Projects	Department	Dean Office

Final Year Project (FYP) Defense Proposal Form

University/Institute Detail:

Name of University/Institution	Institute of Space Technology	Physical Address:	Islamabad Expressway Sector H DHA Phase 2
Telephone & Fax No:	051-9075100	Department Name:	Computing
City:	Islamabad	Province:	Punjab

Nominated Project

Details:-

Project Supervisor Name and Designation:	Sir Faran Mehmood	Contact Details:	Email: faran.mehmood@ist.edu.pk Cell No: 03355504554 Off No:
Project Supervisor Qualification:		No of Publications of Supervisors:	
Students Name(s):	1. <u>Rana M. Areeb</u> 2. <u>M. Hasnain</u>	Students Mobile No:	1. <u>03030538811</u> 2. <u>03347780615</u>
Students CGPA	1. <u>2.76</u> 2. <u>2.74</u>	Students Email:	1. ranaareeb1029@gmail.com 2. m.hasnain1015@gmail.com
Degree Program/ Title:	BS CS	Area of Specialization :	

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Final Year Project Details:

A. Project Title:	Forensic Acoustics for Synthetic Speech Detection.
B. Project Start Date:	12th October,2025
C. Project Finish Date:	12th June,2026
D. Project Summary (less than 200 words)	
The rise of AI-generated synthetic speech poses serious risks to digital security, identity verification, and information trustworthiness. Modern deepfake audio can closely mimic human voices, making it difficult for listeners and traditional systems to differentiate between real and synthetic speech. This project, <i>Forensic Acoustic Synthetic Speech Detection (FASSD)</i>, aims to develop a robust machine learning–based system capable of accurately detecting AI-generated or manipulated speech. The proposed system analyzes acoustic and environmental cues such as spectral features, replay signatures, background noise artifacts, and device-induced distortions. Using the ASVspoof 2021 dataset and additional augmented data, the system extracts LFCC and Log-Mel spectrogram features and trains a Light Convolutional Neural Network (LCNN) to classify audio as real or synthetic. Environmental augmentations (noise, reverberation, codecs) are used to simulate real-world scenarios and enhance model resilience. The final deliverable is a fully functional software system capable of evaluating speech authenticity and detecting deepfake audio with high reliability.	
E. Project Objectives:	
• To develop a machine learning model that classifies speech as bonafide (human) or spoof (AI-generated). • To utilize forensic audio cues—noise patterns, reverberation, channel artifacts—to detect synthetic manipulation. • To identify when a deepfake voice is embedded into real audio recordings. • To detect replayed synthetic speech, where AI-generated audio is played on a device and re-recorded. • To build a robust evaluation pipeline using augmented and real-world testing scenarios. • To deliver a complete software-based system for deepfake speech detection.	

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F. Project Implementation Method			
- Dataset Preparation: Use ASVspoof 2021 (DeepFake & Logical Access tracks) and create additional augmented datasets with MUSAN noise, RIR-based reverberation, and codec distortions. - Feature Extraction: Generate LFCC (20-dimensional) and Log-Mel spectrogram (64-dimensional) features from all audio samples. - Model Development: Implement and train an LCNN-based deep learning model for binary classification (bonafide vs spoof). - Augmentation & Robustness: Apply noise, replay simulation, and compression to strengthen generalization. - Evaluation Pipeline: Measure performance using EER, ROC-AUC, Accuracy, and confusion matrices. - Deployment: Package final model into a software tool capable of evaluating external test audio files.			
G. Key Milestones of the Project with Dates			
S.No.	Elapsed time since start of the project	Milestone	Deliverable
1.	Week 1-4	Literature Review & Scope Finalization	Research summary + approved scope
2.	Week 5-8	Dataset Setup & Feature Extraction	LFCC + Log-Mel features extracted
3.	Week 9-12	Baseline Model Implementation	LCNN baseline model
4.	Week 13-16	Data Augmentation & Robust Feature Set	Augmented datasets + manifest
5.	Week 17-20	Robust Model Training & Validation	Robust LCNN model + evaluation
6.	Week 21-24	Real-World Testing (Replay & Mix Scenarios)	Testing report + analysis
7.	Week 25-28	Final Report, Documentation & Presentation	Complete FYP report + final demo
H. Final Deliverable of the Project:		(Please tick one of the following)	
Hardware System	Software System	HW/SW Integrated System	
Software Simulation results	Comparative Study	Theoretical Design/Architecture	
Simulator Design	Other Please specify		

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I. Please Specify Technical Details of Final Deliverable		
• A trained convolutional neural network-based classifier (LCNN) for synthetic speech detection. • Feature extraction module for LFCC and Log-Mel computation. • Augmentation engine capable of applying noise, reverberation, replay simulation, and codec effects. • HDF5-optimized feature loader for high-speed inference. • A complete evaluation and visualization pipeline (EER, AUC, learning curves). • A user-friendly software interface/script for testing audio files.		
J. Equipment required for making prototype/working model:		(Please indicate in tabular form the required equipment's along with estimated cost)
Item	Description	
Laptop with GPU	RTX 3050 Laptop GPU (already available)	
External SSD	Used for dataset storage and fast I/O	
Headphones	For manual listening tests	
Microphone (optional)	For replay attack recordings	
Software Libraries	Python, PyTorch, Librosa (open-source)	
K. Benefits of the Project		(Please specify Direct/Indirect beneficiaries)
• Helps prevent misuse of AI-generated voices in fraud, impersonation, and misinformation. • Supports forensic investigations by identifying manipulated or replayed audio. • Useful for telecom, banking, and voice authentication systems. • Contributes to national cybersecurity by strengthening voice-based identity verification. • Provides a foundation for future research in synthetic media detection.		

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It is certified that the FYP titled “**Forensic Acoustics for Synthetic Speech Detection.**” has been approved and is being undertaken by the above-mentioned students as their Final Year Project.

It is undertaken that the undersigned have understood the “terms & conditions” of the program, and further reiterate that, if the subject FYP is approved for funding, the disbursed funds shall be utilized as per “terms & conditions” and the undersigned will be liable to reimburse/refund the unutilized amount and other cost not approved by the IST, if any.

It is further undertaken that after utilization of funds, the said “Fund Utilization Report” shall be furnished along with the other required deliverables as and when required by the IST.

Signature of Supervisor: _____

Dr. Khurram Khurshid

Signatures of HoD: _____